

Lab 5: Dynamic routing configurations: RIP, EIGRP, OSPF, BGP

Objectives

- To understand the concept and importance of dynamic routing in computer networks.
- To study the working principles of RIP, EIGRP, OSPF, and BGP routing protocols.
- To analyze the differences between distance-vector, link-state, and path-vector routing protocols.

Theory

Dynamic routing is a networking method where routers automatically exchange routing information and adjust routes based on network topology changes. Unlike static routing, dynamic routing protocols update routing tables automatically, making them suitable for large and complex networks

Routing Information Protocol (RIP)

RIP is a distance-vector routing protocol that uses hop count as its metric. The maximum hop count allowed is 15, which limits its scalability. RIP periodically sends routing updates every 30 seconds.

Key Features:

- Uses hop count as metric (maximum 15 hops).
- Simple configuration.
- Suitable for small networks.
- Slow convergence compared to modern protocols.

Routing Command

On Cisco, RIP on the routers are configured using the following command:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network [network_address]
```

Router 0:

```
Router> enable
Router# configure terminal
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.1.0
```

```
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

Router 1:

```
Router> enable
Router# configure terminal
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.2.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

Enhanced Interior Gateway Routing Protocol (EIGRP)

EIGRP is an advanced distance-vector routing protocol developed by Cisco. It uses a composite metric based on bandwidth, delay, reliability, and load. EIGRP provides fast convergence and supports large enterprise networks.

Key Features:

- Uses DUAL algorithm for fast convergence.
- Supports VLSM and CIDR.
- Efficient bandwidth usage.
- Suitable for medium to large networks.

Routing Command

On Cisco, EIGRP on the routers are configured using the following command:

```
Router(config)# router eigrp [AS_number]
Router(config-router)# network [network_address] [wildcard_mask]
```

Router 0:

```
Router> enable
Router# configure terminal
Router(config)# router eigrp 100
Router(config-router)# network 192.168.1.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

Router 1:

```
Router> enable
Router# configure terminal
Router(config)# router eigrp 100
```

```
Router(config-router)# network 192.168.2.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

Open Shortest Path First (OSPF)

OSPF is a link-state routing protocol that uses the Shortest Path First (SPF) algorithm to calculate the best path. It uses cost as its metric and supports hierarchical network design with areas, making it highly scalable.

Key Features:

- Uses SPF (Dijkstra) algorithm.
- Fast convergence.
- Supports hierarchical design with areas.
- Suitable for large enterprise networks.

Routing Command

On Cisco, OSPF on the routers are configured using the following command:

```
Router(config)# router ospf [process_id]
Router(config-router)# network [network_address] [wildcard_mask] area [area_id]
```

Router 0:

```
Router> enable
Router# configure terminal
Router(config)# router ospf 1
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

Router 1:

```
Router> enable
Router# configure terminal
Router(config)# router ospf 1
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

Border Gateway Protocol (BGP)

BGP is a path-vector routing protocol primarily used between autonomous systems (AS) on the Internet. It selects the best path based on attributes such as AS-path, local preference, and MED.

Key Features:

- Used for inter-domain routing.
- Highly scalable.
- Uses path attributes for route selection.
- Backbone of the Internet.

Routing Command

On Cisco, BGP on the routers are configured using the following command:

```
Router(config)# router bgp [AS_number]
```

```
Router(config-router)# neighbor [neighbor_IP_address] remote-as [neighbor_AS_number]
```

```
Router(config-router)# network [network_address] mask [subnet_mask]
```

Router 0:

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# router bgp 65001
```

```
Router(config-router)# neighbor 10.0.0.2 remote-as 65002
```

```
Router(config-router)# network 192.169.1.0 mask 255.255.255.0
```

Router 1:

```
Router> enable
```

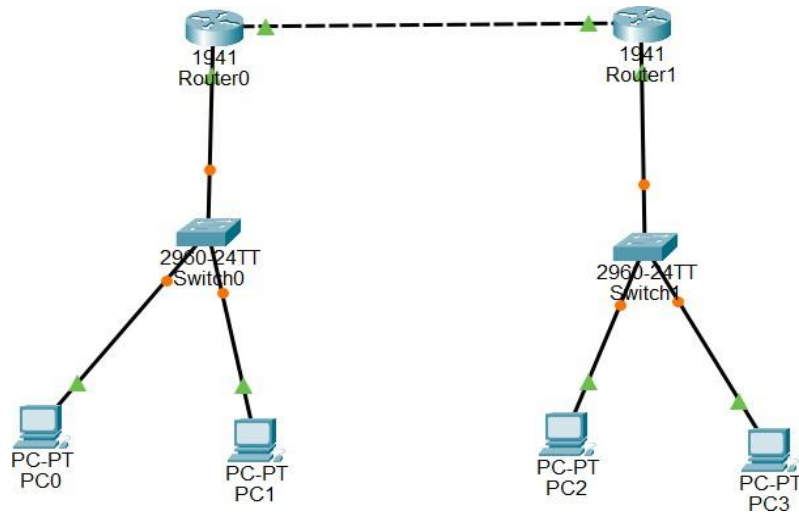
```
Router# configure terminal
```

```
Router(config)# router bgp 65002
```

```
Router(config-router)# neighbor 10.0.0.1 remote-as 65001
```

```
Router(config-router)# network 192.168.2.0 mask 255.255.255.0
```

Network Topology



For PCs:

Device	IPv4 address	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1
PC2	192.168.2.2	255.255.255.0	192.168.2.1
PC3	192.168.2.3	255.255.255.0	192.168.2.1

For Routers:

Device	IPv4 address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
Router 0	192.168.2.0	255.255.255.0	10.0.0.2	gig 0/0	10.0.0.1
				gig 0/1	192.168.1.1
Router 1	192.168.1.0	255.255.255.0	10.0.0.1	gig 0/0	10.0.0.2
				gig 0/1	192.168.2.1

Observations

Ping Result using RIP

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping Result using EIGRP

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping Result using OSPF

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping Result using BGP

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Discussion

Dynamic routing protocols allow routers to automatically share and update routing information when network changes occur. RIP is simple but limited to small networks due to its hop count restriction. EIGRP provides faster convergence and better performance for enterprise networks. OSPF is highly scalable and suitable for large organizations because of its link-state design. BGP is mainly used for routing between autonomous systems and forms the backbone of the Internet.

Conclusion

Dynamic routing improves network efficiency, scalability, and reliability. RIP is suitable for small networks, EIGRP and OSPF are ideal for enterprise environments, and BGP is essential for Internet-level routing. Choosing the right protocol depends on network size and requirements.